FLAT BASE CHANGE CAN DESTROY THE KOSZUL-REGULARITY OF A REGULAR SEQUENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Companion of lemma-strange-regular-sequence, which supplies the ring B and its regular sequence.

Editorial note. The label lemma-base-change-regular-sequence names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The module E and the square-zero extension ring of the section construction, whose properties (a) through (d) are proved in section prose and restated in the companion lemma lemma-strange-regular-sequence
- Over a principal ideal domain a finitely generated torsion-free module is flat; the ring k[z] localized at the prime generated by z is a discrete valuation ring
- More on Algebra, Lemma more-algebra-lemma-regular-koszul-regular: a regular sequence is Koszul-regular; hence a pair whose Koszul homology does not vanish cannot be a regular sequence

Source reference. Original reference 0640: sources/stacks-examples/pinned-original-source.tex, source lines 1333–1333. Prerequisite pointer invoked at 1362; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 062F: sources/stacks-chow/tags.txt, source lines 7794–7794. Prerequisite pointer invoked at 1371; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *There exists a local homomorphism of local rings $A \to B$ and a regular sequence x, y in the maximal ideal of B such that $B/(x, y)$ is flat over A, but such that the images $\overline{x}, \overline{y}$ of x, y in $B/\mathfrak{m}_A B$ do not form a regular sequence, nor even a Koszul-regular sequence.*

Proof. Set $A = k[z]_{(z)}$ and let $B = (k[x, y, z] \oplus E)_{(x,y,z,E)}$. Since x, y, z is a regular sequence in B, see proof of Lemma 0640, we see that x, y is a regular sequence in B and that $B/(x, y)$ is a torsion free A-module, hence flat. On the other hand, there exists a nonzero element $\delta \in B/\mathfrak{m}_A B = B/zB$ which is annihilated by $\overline{x}, \overline{y}$. Hence $H_2(K_\bullet(B/\mathfrak{m}_A B, \overline{x}, \overline{y})) \neq 0$. Thus $\overline{x}, \overline{y}$ is not Koszul-regular, in particular it is not a regular sequence, see More on Algebra, Lemma 062F. □

Date: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.